Yating Wu

Ph.D., UT Austin ECE

yating.wu@utexas.edu
lingchensanwen.github.io

Impact Highlights

Major funded AI roles I contribute question-based discourse modeling, evidence grounding, and evaluation methods to major federally funded AI initiatives: CosmicAI, a $20M NSF-Simons AI Institute for Cosmic Origins, supported by NSF Cooperative Agreement 2421782 ($10,000,000 intended NSF award; $4,723,958 awarded to date) and Simons Foundation grant MPS-AI-00010515; IFML, the NSF AI Institute for Foundations of Machine Learning, supported by NSF CCF-2019844 ($19,800,000 awarded; $20,000,000 estimated total); and NSF CAREER IIS-2145479 ($540,543). Across these programs, I turn broad AI reliability and accessibility goals into concrete datasets, benchmarks, evaluation protocols, and research outputs.

Funding contributions I contribute to the following federal, foundation, and industry funding streams through research outputs, datasets, benchmarks, and project reporting: NSF IIS-2145479, IIS-2145280, AF-1901292, CNS-2148141, TRIPODS CCF-1934932, IFML CCF-2019844, IIS-1850153, and IIS-2107524; Open Philanthropy; Western Digital, Amazon, Cisco, WNCG IAP, UT Austin Machine Learning Lab (MLL), Wipro, the Knight Foundation, the Micron Foundation, Good Systems, the Texas Advanced Computing Center (TACC), and the Stanly P. Finch Centennial Professorship in Engineering.

NSF workstream lead I lead the Questions Under Discussion (QUD) workstream in *Discourse Processing and Content Generation for Document Simplification* under NSF CAREER IIS-2145479 and have prepared the annual NSF progress-reporting materials to date.

Award-winning work Led QSalience, an EMNLP 2024 Oral paper selected for an Outstanding Paper Award (20 of 6,105 reviewed submissions, top 0.33%); selected as an MIT EECS Rising Star, recognizing emerging researchers whose work shows high impact and strong potential to shape the future of computing and AI.

Independent adoption Seven peer-reviewed publications and two current preprints in NLP, AI/ML, and multimodal scientific reasoning; Google Scholar: 95 citations, h-index 6, i10-index 6; Semantic Scholar: 92 citations, h-index 6, and 12 highly influential citations; independent groups have built on my QUD parser, QUDeval, and QSalience in EMNLP, ACL, SIGDIAL, and TACL work (May 2026).

Education

2022 – 2026 **Ph.D. in Electrical and Computer Engineering**, *The University of Texas at Austin*
Advisors: Junyi Jessy Li, Alexandros G. Dimakis

2020 – 2022 **M.S. in Electrical and Computer Engineering**, *The University of Texas at Austin*

2014 – 2019 **B.Eng. in Computer Science & B.A. in Japanese**, *Dalian University of Technology*

2017 - 2018 **Exchange student in Computer Science**, *The University of Tokyo*
Advisor: Toshihiko Yamasaki

Research Overview

I develop question-based methods for reliable, evidence-grounded language models.

- Model discourse and implicit reasoning in long documents.
- Build grounding, salience, and coherence benchmarks.
- Apply to accessibility, long-context agents, and scientific QA.

Industry Experiences

Jun - Oct. 2025 **Applied Scientist Intern**, *AWS AI Labs*, Santa Clara, CA

- Long-context memory support for LLM coding agents

May - Aug. 2024 **Applied Scientist Intern**, *Amazon Alexa AGI*, Sunnyvale, CA

- Enhancing planning and tool-use capabilities in LLM agents

Work In Progress and Preprints

[1] **Yating Wu**, William Rudman, Venkata S Govindarajan, Alexandros G. Dimakis, Junyi Jessy Li "Multimodal QUD: Inquisitive Questions from Scientific Figures." [Under Review; arXiv]

[2] **Yating Wu**, Yuhao Zhang, Sayan Ghosh, Sourya Basu, Anoop Deoras, Jun Huan, Gaurav Gupta "ContextWeaver: Selective and Dependency-Structured Memory Construction for LLM Agents." [Under Review; arXiv]

Publications

[3] Ramya Namuduri, **Yating Wu**, Anshun Asher Zheng, Manya Wadhwa, Greg Durrett and Junyi Jessy Li "QUDsim: Quantifying Discourse Similarities in LLM-Generated Text." Conference on Language Modeling **(COLM)**, 2025 [Paper]

[4] **Yating Wu***, Ritika Rajesh Mangla*, Alexandros G. Dimakis, Greg Durrett, Junyi Jessy Li. "Which questions should I answer? Salience Prediction of Inquisitive Questions." Conference on Empirical Methods in Natural Language Processing **(EMNLP Oral)**, 2024. **Outstanding Paper Award (Top 0.33%).** [Paper]

[5] Negin Raoof*, **Yating Wu***, Carlos Bonilla*, Junyi Jessy Li, Stephanie M Grasso, Alex Dimakis, Zoi Gkalitsiou. "Modeling Bilingual Disfluencies with Large Language Models." Workshop on LLMs and Cognition in International Conference on Machine Learning **(ICML Workshop)**, 2024. [Paper]

[6] **Yating Wu**, Ritika Rajesh Mangla, Greg Durrett, Junyi Jessy Li. "QUDeval: The Evaluation of Questions Under Discussion Discourse Parsing." Conference on Empirical Methods in Natural Language Processing **(EMNLP Oral)**, 2023. [Paper]

[7] **Yating Wu***, William Sheffield*, Kyle Mahowald, and Junyi Jessy Li. "Elaborative Simplification as Implicit Questions Under Discussion." Conference on Empirical Methods in Natural Language Processing **(EMNLP)**, 2023. [Paper]

[8] Wei-Jen Ko, **Yating Wu**, Cutter Dalton, Dananjay Srinivas, Greg Durrett and Junyi Jessy Li. "Discourse Analysis via Questions and Answers: Parsing Dependency Structures of Questions Under Discussion." Findings of the Association for Computational Linguistics **(ACL)**, 2023. [Paper]

[9] Venelin Kovatchev, Trina Chatterjee, Venkata S Govindarajan, Jifan Chen, Eunsol Choi, Gabriella Chronis, Anubrata Das, Katrin Erk, Matthew Lease, Junyi Jessy Li, **Yating Wu**, Kyle Mahowald. "longhorns at DADC 2022: How many linguists does it take to fool a Question Answering model? A systematic approach to adversarial attacks." DADC Workshop in The Nations of the Americas Chapter of the Association for Computational Linguistics **(NAACL Workshop)**, 2022 [Paper]

Honors and Fellowships

Honors

2025 **EECS Rising Stars, MIT**

2024 **Outstanding Paper Award, EMNLP 2024**

2021 **1st Place in VMware Codehouse Palo Alto**

2019 **Outstanding Graduates of Dalian University of Technology**

Fellowships Under Review

2026	**Microsoft Research Fellowship**, Microsoft Research
2026	**Bloomberg Data Science PhD Fellowship**, Bloomberg

Fellowships

2025	**Outstanding Student Lecture Series Fellowship, UT Austin**
2023-2024	**Professional Development Awards, UT Austin**
2022-2025	**Head TA fellowship, UT Austin**

Teaching Experience

Head Teaching Assistant

CS391L **Machine Learning (Graduate)**, Summer 2022 – Fall 2025
Instructors: Adam Klivans, Qiang Liu, UT Austin

Teaching Assistant

CS391L **Machine Learning (Graduate)**, Fall 2021 – Spring 2022
Instructors: Adam Klivans, Qiang Liu, UT Austin

Recitation Instructor

EE422C **Software Design and Implementation II (Undergraduate)**, Spring 2021
Independently designed and taught weekly recitation sections

EE422C **Software Design and Implementation II (Undergraduate)**, Summer 2020
Independently designed and taught weekly recitation sections

Mentoring Experience

Master's student	**Ritika Mangla**, 2022-2024, co-authored paper [1] and [3], UT Austin
Master's student	**Ashwin Ram**, 2024, UT Austin

Presentations

Talks

2025	Which Questions Should I Answer? Salience Prediction of Inquisitive Questions, Outstanding Student Lecture Series, UT Austin
2024	Which Questions Should I Answer? Salience Prediction of Inquisitive Questions, EMNLP 2024
2023	QUDeval: The Evaluation of Questions Under Discussion Discourse Parsing, EMNLP 2023

Poster Presentations

2024	Elaborative Simplification as Implicit Questions Under Discussion, MLL Research Symposium, UT Austin
2023	Elaborative Simplification as Implicit Questions Under Discussion, EMNLP 2023
2022	Discourse Analysis via Questions and Answers: Parsing Dependency Structures of Questions Under Discussion, ACL 2022

Industrial Presentation

2025	ContextWeaver: Selective and Interpretable Long Context Construction for LLM Agents, AWS AI Labs
2024	Self Adaptive Question Generation for Planning Agent, Amazon AGI

Service

Conference Reviewer

ACL Rolling Review (ARR) 2024-2025, ACL 2024-2025
EMNLP 2024
COLM 2024-2025
Workshops: ICML 2025 XAI4Science, AAAI 2025 MARW, NeurIPS 2024 Behavioral ML, ICML 2024 DMLR, ICML 2024 LLMs and Cognition

Department Service

Graduate student representative on faculty hiring panel

TA representative on online course infrastructure building